TEXORA · LECTURE SHEET

Sewing Machinery

The factory fleet classified by bed, stitch and automation: lockstitch, overlock, coverstitch, feed-off-arm, buttonhole, bartack and blind-stitch machines, their feed systems, needles and capacity planning.

TE-3107 · APPAREL MANUFACTURING TECHNOLOGY-II · TOPIC 2 · LEVEL 3 TERM 1

1. How sewing machines are classified

Industrial sewing machines are classified three ways at once: by **bed shape**, by **stitch type formed**, and by **degree of automation**. A specification such as 'single-needle, flat-bed, lockstitch, with thread trimmer' uses all three.

BED	FORM AND USE
Flat bed	Flat work surface level with the table; the general machine for shirts, trousers, flat parts.
Cylinder bed	Narrow horizontal arm; tubular parts — sleeves, trouser legs, cuffs — slip over the cylinder.
Post bed	Vertical post carries the needle; three-dimensional work such as shoes, bags, caps.
Off-the-arm (feed-off-arm)	Arm parallel to the operator edge; lap-felling chainstitch machines for shirt and jeans side seams.

By automation: plain mechanical machines; machines with **automatic thread trimming, back-tack and presser-foot lift** (the modern factory standard, driven by a servo motor); **semi-automatic workstations** that perform one fixed cycle (buttonhole, button attach, bartack, pocket setting); and **automatic units** with templates or CNC pattern sewing for labels, decorative panels and small parts.

2. The principal machine types of a garment factory

MACHINE	STITCH	OPERATIONS IN SHIRT/TROUSER WORK
Single-needle lockstitch (SNLS)	301	General seaming, topstitching, collar, placket, pocket setting.

Double-needle lockstitch	301 ×2	Parallel topstitch rows: trouser outseams, shirt front plackets, yokes.
Overlock (3/4/5-thread)	504/514/516	Edge neatening, knit seaming, trouser serging, safety-stitch side seams.
Flatlock / coverstitch	602–607	Knit hems, flat seams, binding and decorative covering.
Feed-off-arm chainstitch	401	Lap-felled side seams of shirts and jeans.
Buttonhole (lockstitch or chain)	304/101-based	Straight and eyelet keyhole buttonholes, cut before or after sewing.
Button attach	101/304	Sewing flat and shank buttons with set stitch counts and cross-stitch option.
Bartack	304-pattern	Reinforcing pocket corners, fly ends, belt loops (typically 28 or 42 stitches per tack).
Zigzag	304	Elastic attachment, appliqué, bartack-type reinforcement on light work.
Blind stitch	103	Invisible hemming of trouser bottoms and skirts.

3. Machine anatomy: the parts that matter

Every lockstitch machine, whatever its badge, contains the same functional organs, and Topic 9 of the sessional course examines their thread path and adjustment in the hand. The **needle bar** drives the needle; the **take-up lever** meters thread to the stitch; the **rotating hook** (oscillating in older designs) carries the loop around the bobbin; the **feed dog and presser foot** transport the fabric; the **stitch-length regulator** (dial or, on automatics, a panel setting) sets feed throw; the **tension assembly** — two discs and a spring — sets needle-thread tension against the bobbin-case spring below. Speeds are quoted in stitches per minute (SPM): modern SNLS machines run to 5,000–5,500 SPM, overlocks to 6,500–7,500 SPM — but a factory sets working speed by operation and material, not by the brochure maximum.

4. Feed systems compared

FEED SYSTEM	HOW IT WORKS	WHERE IT IS NEEDED
Drop feed	Feed dog only; upper ply dragged by the foot.	Light, short, stable seams.
Needle feed	Needle moves with the fabric while in it, plus drop feed.	Slippery or shifting plies; precise matching of checks.

Walking foot (triple/unison feed)	Alternating feet walk with the feed dog and the needle feed together.	Heavy and multi-ply work: waistbands, jeans, upholstery.
Differential feed	Two feed dogs with adjustable ratio (overlock/flatlock).	Controls knit stretch: gather (>1) or stretch out (<1) to prevent wavy or fluted edges.
Puller / belt feed	Driven roller or belt behind the needle assists transport.	Very long seams and heavy bundles.

5. Needles: system, point and size

A needle is specified by **system** (e.g. DB×1 for SNLS, DC×27 for overlock), **point** and **size**. Sizes are given in the metric Nm number (blade diameter in hundredths of a mm: Nm 75 = 0.75 mm) with the Singer number alongside (Nm 75 ≈ Singer 11).

FABRIC	POINT	NEEDLE NM (SINGER)	TYPICAL THREAD (TOPIC 3)
Fine shirting, voile	Sharp / light ball	65–75 (9–11)	Fine spun or core-spun, tex 18–24
Medium shirting, poplin	Sharp	75–90 (11–14)	Tex 24–30
Denim, twill trousers	Sharp / heavy set	90–110 (14–18)	Tex 40–60 core-spun
Knits (jersey, interlock)	Ball point (SES/SUK)	70–90 (10–14)	Textured or spun, tex 18–30
Leather and coated	Cutting point	90–120 (14–19)	Heavy bonded/core-spun

The **thread must pass freely through the needle eye**: as a rule the thread should fill roughly 40% of the eye so it seats in the groove without shredding. Too large a needle in fine fabric leaves holes and invites slippage; too fine a needle with heavy thread deflects, heats and breaks.

6. Choosing machines for a line: a worked layout question

Example. A trouser line targets 600 pieces/day. Operation 'serge trouser parts' takes 0.55 min/piece on a 5-thread overlock. One machine at 85% efficiency in a 480-min day delivers 480 × 0.85 ÷ 0.55 = 742 pieces — one machine suffices on capacity, but the factory still stations **two**, because a single breakdown would starve every downstream operation. Machine planning always combines the capacity

calculation with this redundancy judgement, and with the stitch specification: no capacity figure can convert a 301 operation into a 401 operation if the buyer's spec forbids it.

Glossary

TERM	MEANING
SPM	Stitches per minute; the machine speed measure used in capacity planning.
Unison (triple) feed	Combined needle feed, walking foot and feed dog moving together while the needle is in the material.
Differential feed ratio	The speed ratio of the two feed dogs of an overlock or coverstitch machine; above 1 gathers, below 1 stretches.
Needle system	The manufacturer's dimensional code (shank, length, eye position) that decides which needles fit a machine class.
Bartack	A dense block of zigzag/lockstitches reinforcing a stress point, specified by stitch count and dimensions.

Key points and summary

- Classify any machine three ways: bed, stitch type, automation level.
- The factory's core fleet: SNLS, overlock, coverstitch, feed-off-arm, buttonhole, button attach, bartack, blind stitch.
- Feed system is chosen by material behaviour: drop feed for stable work, unison feed for heavy/shifting plies, differential feed to control knits.
- Needle size and point are matched to fabric and thread together; the thread must run freely in the needle eye.

Practice

Answer in your own words; brief model answers follow.

1. A flatlock machine is described as 'cylinder-bed, type 607'. Decode both parts of the description.
2. An operation takes 0.80 min/piece; the target is 900 pieces in a 480-min day at 80% efficiency. How many machines are required? Show the working.
3. Why does a knit hemming operation use differential feed set below 1 when the hem comes out wavy?

4. Choose needle system family, point and Nm for (a) sewing a denim jeans inseam, (b) hemming a cotton single-jersey T-shirt.
5. State two operations on a basic trouser that specifically require a bartack machine and why a lockstitch machine cannot substitute.

Answer key

1. Cylinder bed: the work slips over a narrow arm, suiting tubular parts. Type 607: ISO 4915 class 600 covering chainstitch with cover thread — a flatlock/coverstitch formation.
2. Capacity per machine = (480 × 0.80) ÷ 0.80 = 480 pieces. Machines required = 900 ÷ 480 = 1.875, so 2 machines.
3. A wavy hem means the edge was stretched during sewing; a differential ratio below 1 feeds the back dog slower, letting the fabric relax/shorten so the edge lies flat.
4. (a) SNLS/chain machine needle, sharp or heavy-set point, Nm 100–110 with tex 40–60 thread. (b) Ball-point (SES) Nm 75–80 so the point parts the loops instead of piercing them.
5. Pocket-corner and belt-loop (and fly-end) reinforcement: the bartack lays a programmed dense stitch block (e.g. 42 stitches) in a fixed rectangle that locks the stress point; a plain lockstitch machine has no pattern mechanism to form that block repeatably.

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